

7-1 Project Three

1. Operating Platform

For The Gaming Room's Draw It or Lose It application, the most appropriate operating platform is a Linux-based cloud environment. Linux is widely used for web-based applications due to its reliability, scalability, and low cost. Since the application needs to support multiple users across different platforms, a cloud-based Linux server allows the system to scale easily as demand increases. This approach also reduces the need for physical hardware and allows the company to pay only for the resources it uses.

2. Operating Systems Architectures

The recommended architecture for this application is a distributed, client-server model using a Linux-based server. In this setup, the server handles the game logic, data processing, and communication, while clients access the application through web browsers on different devices. Linux supports this architecture well because of its efficient kernel design, which manages processes, threads, and system resources effectively. This allows the system to handle high workloads and multiple users simultaneously. This architecture also ensures the application remains platform-independent, allowing users on Windows, Mac, or mobile devices to access the same system without compatibility issues.

3. Storage Management

For storage management, a cloud-based storage solution combined with a database system is recommended. The application requires storing a large number of image files, as well as user data, teams, and game history. Cloud storage services such as AWS S3 or similar platforms provide scalable and reliable storage for image files, while a relational database can be used to manage structured data like users and game sessions. This setup allows the system to expand storage capacity as needed while maintaining fast and efficient access to data. It also supports redundancy and backup systems to prevent data loss.

4. Memory Management

Memory management is critical for ensuring smooth gameplay, especially since images must be rendered quickly during each round. The recommended approach is to load only the necessary images into memory during gameplay rather than storing the entire image library at once. This reduces memory usage and improves performance. Techniques such as caching frequently used data and using efficient data structures can also help optimize memory usage. Additionally, Linux uses virtual memory and paging techniques to manage memory efficiently, allowing the system to handle multiple users and processes without running out of physical memory.

5. Distributed Systems and Networks

Draw It or Lose It will rely on a distributed system where clients communicate with the server over a network using RESTful APIs. This allows users on different devices and operating systems to interact with the same application. The network must be reliable and capable of handling high traffic, especially during peak usage. Load balancing can be used to distribute requests across multiple servers, improving performance and preventing overload. It is also important to consider potential issues such as network latency or outages, so implementing redundancy and failover systems will help maintain availability.

6. Security

Security is essential for protecting user data and ensuring the integrity of the application. The system should use HTTPS to encrypt data transmitted between the client and server.

Authentication and authorization mechanisms, such as role-based access control, should be implemented to ensure users can only access appropriate resources. Additionally, input validation should be used to prevent attacks such as injection attacks. Regular updates and security patches are also important to protect against vulnerabilities. By combining these security measures, the application can provide a safe and reliable experience for users across all platforms.